

Data Converters

Exercise 01 – Sampling, Aliasing, Reconstruction, DNL, INL

Task I – Aliasing & Sampling Frequency

Consider the analog signal $x(t) = \cos(2\pi f_1 t) + \cos(2\pi f_2 t) + \cos(2\pi f_3 t)$ with $f_1 = 0.5$ MHz, $f_2 = 3$ MHz, and $f_3 = 6$ MHz.

- a. What is the minimum sampling frequency f_s that avoids aliasing?

$$f_s \geq 12 \text{ MHz}$$

- b. Assume that we sample $x(t)$ at $f_s = 4$ MHz. What is the discrete-time signal we obtain after sampling?

$$x[n] = \cos(0.25\pi \cdot n) + \cos(0.5\pi \cdot n) + \cos(\pi \cdot n)$$

- c. State two other signals $x'(t)$ that have the same discrete-time representation as $x(t)$ in Tl.b.

$$x_1(t) = \cos(2\pi \cdot 0.5 \text{ MHz} \cdot t) + \cos(2\pi \cdot 1 \text{ MHz} \cdot t) + \cos(2\pi \cdot 2 \text{ MHz} \cdot t)$$

$$x_2(t) = \cos(2\pi \cdot 0.5 \text{ MHz} \cdot t) + \cos(2\pi \cdot 7 \text{ MHz} \cdot t) + \cos(2\pi \cdot 10 \text{ MHz} \cdot t)$$

Task II – Anti-Aliasing Filtering

For the system shown in Figure 1, what is the minimum sampling frequency f_s that guarantees more than -50 dB attenuation of in-band aliasing components in the discrete-time signal $y[n]$? Note that the signal bandwidth f_b is 1 MHz.

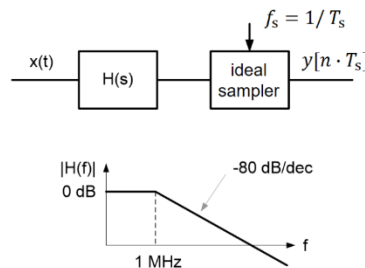


Figure 1: Anti-aliasing filtering.

$$f_s \approx 5.22 \text{ MHz}$$

Task III – Reconstruction

A discrete-time signal $x[n]$ with a sampling rate $f_s = 250$ MHz is converted to a continuous-time representation using zero-order hold pulses.

- Plot the amplitude envelope of the continuous-time signal resulting from hold pulse widths of $T_{p1} = 1$ ns, $T_{p2} = 2$ ns, and $T_{p3} = 4$ ns all on the same graph up to 1 GHz. Leave the x-axis in absolute frequency rather than normalizing it to f_s .
- Suppose that the discrete-time signal $x[n]$ represents a 60 MHz sine wave with amplitude $\hat{u}$. For the 5 ns hold case, calculate the amplitudes (normalized to $\hat{u}$) of all tones present in the output spectrum up to 500 MHz.

$$f = 60 \text{ MHz} \rightarrow \text{envelope magnitude: } 0.9079$$

$$f = 190 \text{ MHz} \rightarrow \text{envelope magnitude: } 0.2867$$

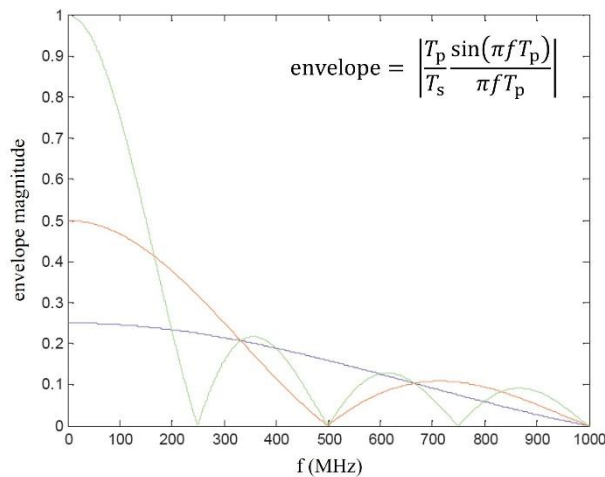
$$f = 310 \text{ MHz} \rightarrow \text{envelope magnitude: } 0.1757$$

$$f = 440 \text{ MHz} \rightarrow \text{envelope magnitude: } 0.1238$$

Task IV – DNL & INL (ADC)

Consider a monotonic ADC with output codes 0, 1, 2, ..., M.

- Show that $INL(j) = \sum_{i=1}^{j-1} DNL(i)$.



- Use (a) to verify that $\sum_{i=1}^{M-1} DNL(i) = 0$.

$$\sum_{i=1}^{M-1} DNL(i) = INL(M) = 0$$

- You report your measurement results of your 12-bit prototype ADC to Dr. Keller:
 - The SQNR (measured using a full-scale sine wave) seems to be excellent, at least 80 dB.

$$12\text{-bit ADC} \rightarrow SQNR_{\max} = 6.02 \text{ dB} \cdot B + 1.76 \text{ dB} = 74 \text{ dB}$$

Please turn the page.

- For one of your chips, the DNL is not so great, i.e., $DNL = 0.1/-2.5 \text{ LSB}$.
 $DNL = -1 \rightarrow$ missing code, i.e., $DNL < -1$ is not feasible.
- For one chip, you measured $DNL = 0.4/-0.4 \text{ LSB}$ and $INL = 0.1/-0.1 \text{ LSB}$.
If $DNL = 0.4$, then two subsequent INL values must differ by 0.4.
This is not feasible if the INL is bounded by ± 0.1 .

Dr. Keller shakes his head in disbelief... Why?

Task V – INL (DAC)

Consider a 12-bit DAC with a DNL curve following a sine wave characteristic with $DNL = 0$ at the endpoints and the center code as shown in Figure 2. The maximum DNL values are ± 0.005 .

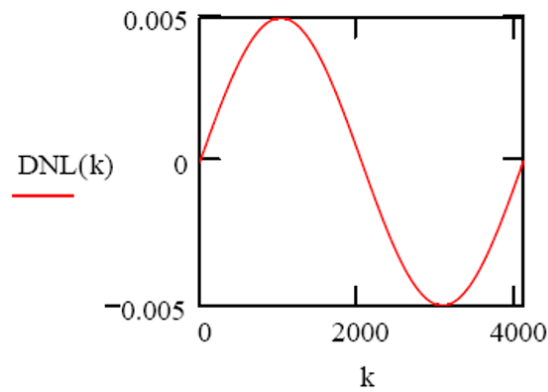


Figure 2: DNL of a 12-bit ADC.

a. What is the peak INL of this DAC?

$$INL_{\max} \approx 6.517$$

b. At which code does the peak INL occur?

$$\text{at mid-code, i.e., } k = 2047 \text{ or } k = 2048$$

c. In which code interval(s) is the INL less than 1 LSB?

$$k < 525 \text{ or } k > 3570$$